AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2019

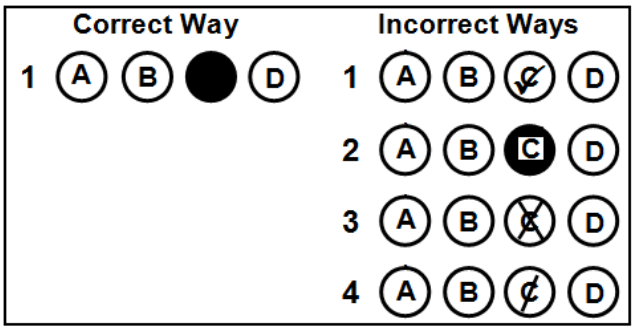
Biology Paper I

Time: 50 minutes Marks: 35

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 35 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. Zara is interested in learning about the global environmental change and its effects on terrestrial and aquatic ecosystems.

   The field of biology that relates to Zara's interest the MOST is

   A. sociobiology.
   B. bioclimatology.
   C. marine biology.
   D. fresh water biology.

2. The given statements represent an example of deductive reasoning.

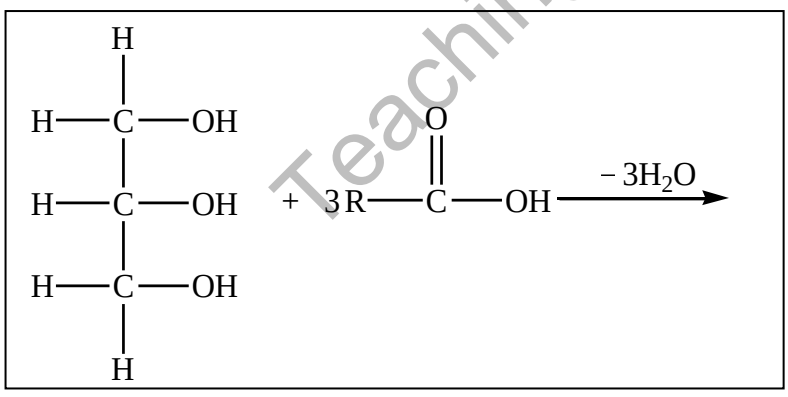
   I. Red meat contains iron.
   II. Beef is red meat.
   III. Beef contains iron.

   The statement(s) that indicate(s) the specific conclusion of the deductive reasoning is

   A. II only.
   B. III only.
   C. I and II.
   D. I and III.

3. The property of water molecule which results in hydrogen bonding is

   A. polarity.
   B. cohesion.
   C. heat of vapourisation.
   D. specific heat capacity.

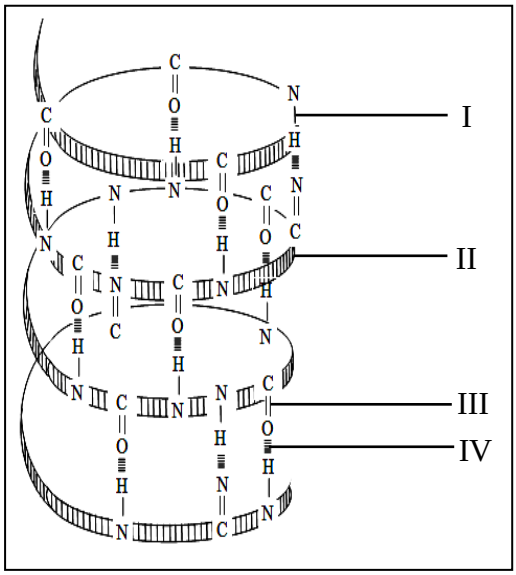
4. 

   On condensation, the given reactants will yield a molecule of

   A. steroid.
   B. carotenoid.
   C. acylglycerol.
   D. phospholipid.

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5. The component of the given α-helical structure of protein which maintains its stability is



A. I.
B. II.
C. III.
D. IV.

6. The given table shows the percentage of product formation in an enzyme catalysed reaction at different temperatures.

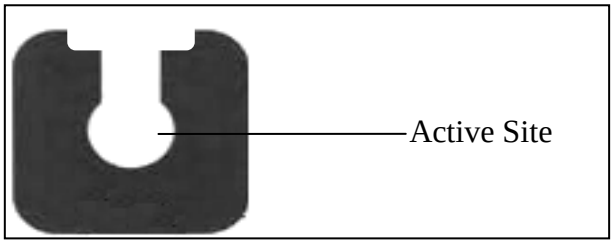
S. No.	Temperature °C	Percentage (%) of Product Formation
I	20	40
II	25	60
III	30	80
IV	35	100

The optimum temperature for the enzyme to form the product would be

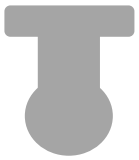
A. I.
B. II.
C. III.
D. IV.

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7. The given diagram represents an enzyme.



With reference to the lock and key model, the competitive inhibitor for the given enzyme would be

			
A	B	C	D

8. Centrifugation is a technique that involves the use of centrifugal force for the separation of

A. bacteria on the basis of Gram staining.
B. organelles on the basis of their density.
C. biological molecules on the basis of their polarity.
D. chlorophyll pigments on the basis of their ability to absorb light.

9. Human pancreatic cells synthesise digestive enzymes in large quantity due to the presence of a large number of

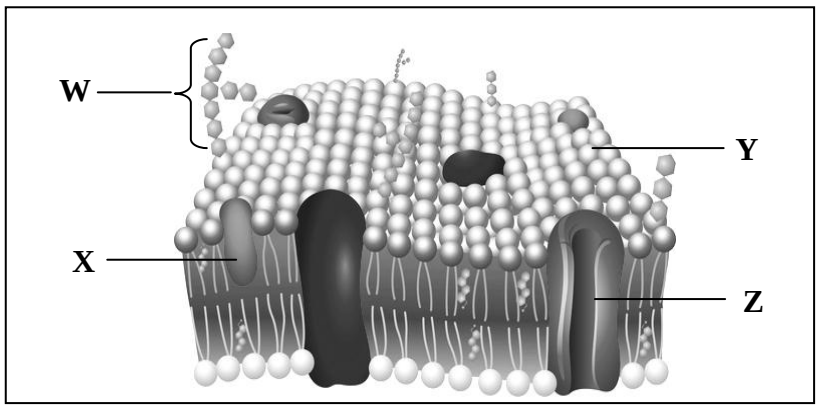
A. ribosomes.
B. lysosomes.
C. Golgi bodies.
D. smooth endoplasmic reticulum.

10. The presence of deoxyribonucleic acid **(**DNA) and ribosomes in the mitochondria indicates that mitochondria are capable of forming

A. proteins.
B. glycolipids.
C. carbohydrates.
D. adenosine triphosphate (ATP).

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11. The given diagram shows the structure of cell membrane.



Which of the labelled components, if removed from the cell membrane, will limit the movement of large and charged particles across the cell membrane?

A. W
B. X
C. Y
D. Z

12. During the lytic cycle, the function of lysozymes released by phage is to

A. synthesise viral genome.
B. dissolve bacterial cell wall.
C. break down bacterial DNA.
D. speed up the formation of proteins.

13. Transduction is a type of genetic recombination in which DNA transfers from one bacterium to another through

A. viruses.
B. ingestion.
C. binary fission.
D. physical contact.

14. The groups of protists that act as the major producers of aquatic ecosystems are

A. diatoms and dinoflagellates.
B. green algae and zooflagellates.
C. foraminiferans and euglenoids.
D. brown algae and water moulds.

15. In contrast to Kingdoms Animalia and Plantae, the organisms of Kingdom Fungi have

A. a cell wall.
B. nuclear mitosis.
C. centrioles in cells.
D. heterotrophic mode of nutrition.

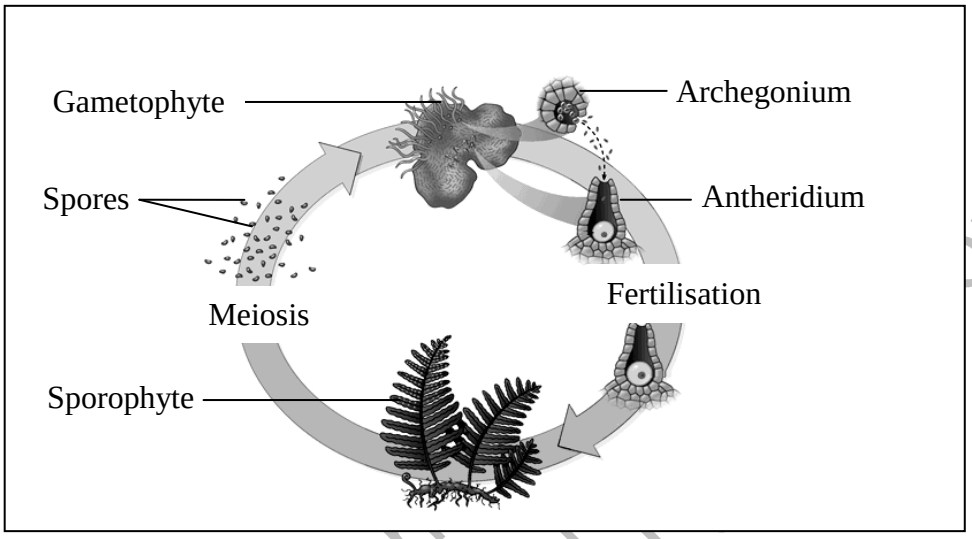
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16. Maria examines a culture of hyphae of an unknown fungal species. She observes the presence of non-septate, multinucleated hyphae and spores formed in sporangia.

Based on her observations, the unknown species belongs to the phylum

A. Ascomycota.
B. Zygomycota.
C. Basidiomycota.
D. Deuteromycota.

17. The given diagram shows the life cycle of a plant.

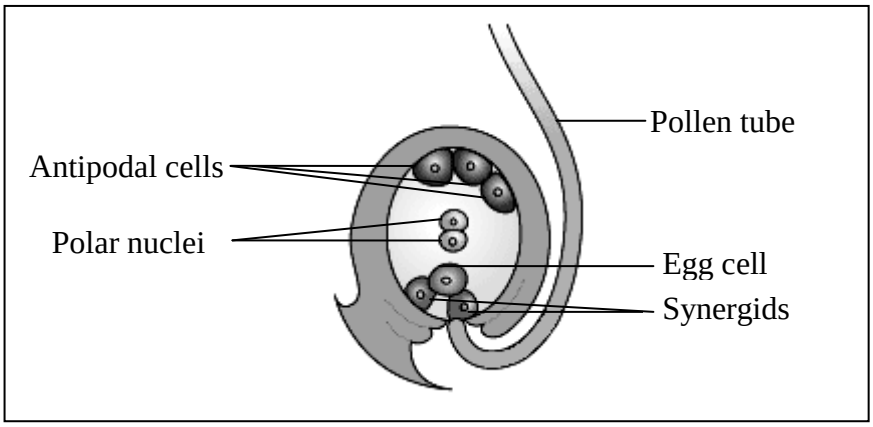


The feature in the given life cycle due to which this plant is classified as a tracheophyte is the presence of

A. haploid spores.
B. dominant sporophyte.
C. independent gametophyte.
D. antheridium and archegonium.

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18. The given diagram shows an unfertilised angiosperm ovary.



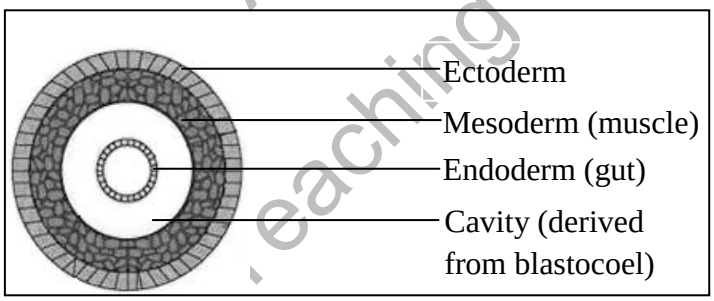
What change would take place in the ovary after double fertilisation?

A. Pollen tube will disappear.
B. Antipodal cells will double in number.
C. Polar nuclei will form triploid endosperm.
D. Egg cell will fuse with a synergid to form diploid zygote.

19. A flower with multiple free perianth leaves [$P_{\text{No. of perianth leaves}}$] is termed as

A. polyandrous.
B. polypetalous.
C. polysepalous.
D. polyphyllous.

20. The given diagram represents body tissue organisation of a



A. triploblastic coelomate.
B. diploblastic acoelomate.
C. triploblastic acoelomate.
D. triploblastic pseudocoelomate.

21. In an experiment, a cell was isolated from an unknown embryo at four-cell stage and allowed to divide. Even after repeated divisions, this isolated cell did not develop into a complete animal.

The embryo used in the above experiment could be of a

A. frog.
B. snail.
C. star fish.
D. sparrow.

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22. The CORRECT sequence of stages (from **I** to **V**) in the life cycle of a cnidarian which shows alternation of generation is

Stage	I	II	III	IV	V
A	polyp	medusa	egg	planula larva	polyp
B	polyp	planula larva	egg	medusa	polyp
C	medusa	planula larva	polyp	egg	medusa
D	medusa	polyp	planula larva	egg	medusa

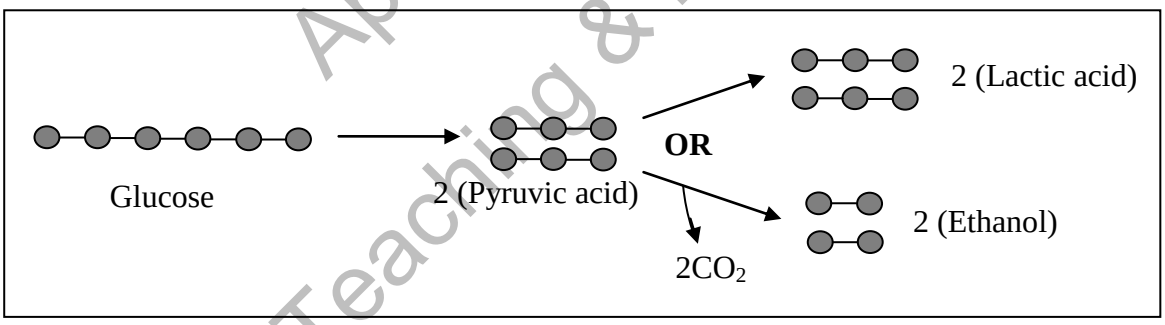
23. Closed circulatory system is present in animals belonging to phylum

A. annelida.
B. mollusca.
C. nematoda.
D. arthropda.

24. In photosynthesis, oxygen molecules are formed as a by-product during

A. photolysis.
B. chemiosmosis.
C. carbon fixation.
D. cyclic phosphorylation.

25. The given diagram summarises the reactions of anaerobic respiration.



Based on the given summary, adenosine triphosphate (ATP) molecules would have been released during the

A. breakdown of pyruvic acid into ethanol.
B. formation of pyruvic acid from glucose.
C. breakdown of pyruvic acid into lactic acid.
D. release of carbon dioxide gas from pyruvic acid.

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26. In cellular respiration, the molecules of nicotinamide adenine dinucleotide (NADH) and Flavin adenine dinucleotide (FADH) are produced during

A. glycolysis.
B. Krebs cycle.
C. pyruvate oxidation.
D. electron transport chain.

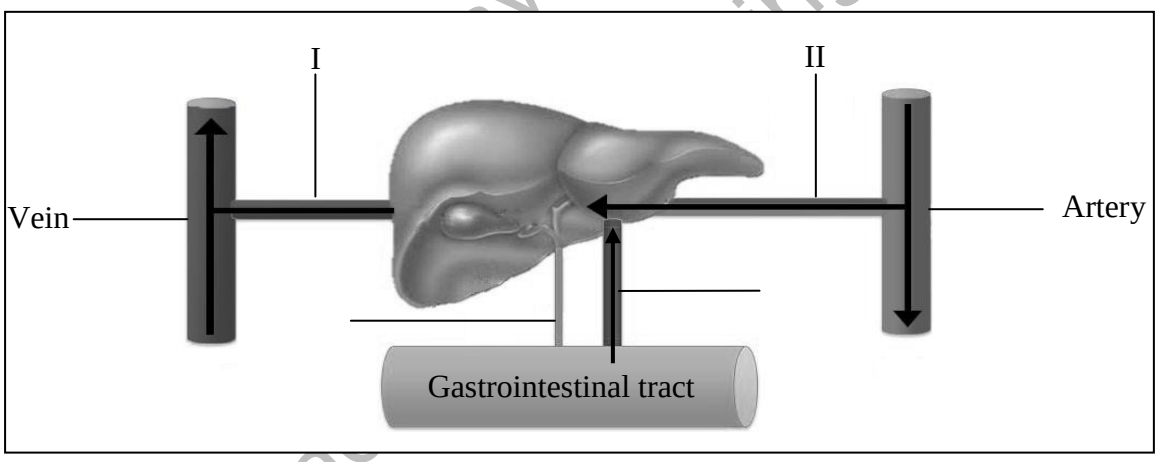
27.

- Its walls are composed of muscles.
- It is present at the posterior end of oral cavity.
- It connects mouth with the oesophagus.

The given features characterises

A. larynx.
B. glottis.
C. pharynx.
D. epiglottis.

28. The given diagram represents hepatic portal system.



The labelled part which contains the largest amount of nutrients is

A. I.
B. II.
C. III.
D. IV.

29. Skin is kept moist by the secretion of epidermal mucous gland cells and coelomic fluid.

The given characteristic feature is a common respiratory adaptation in

A. frog.
B. hydra.
C. cockroach.
D. earthworm.

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30. During inhalation, the pressure and volume of thoracic cavity

	Pressure in Thoracic Cavity	**Volume of Thoracic Cavity**
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

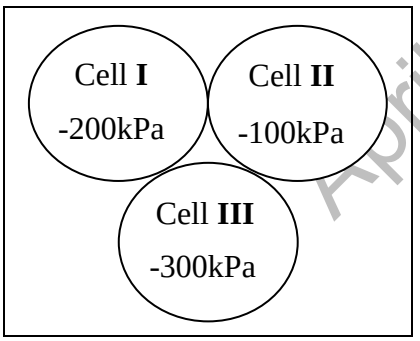
31. What would be the site of the given reaction if it proceeds from right to left?

$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO^-_3$

A. Lungs
B. Trachea
C. Body tissue
D. Diaphragm

32. The given diagram illustrates three adjacent cells with different water potential.

(**Note:** kPa = water potential of cell)

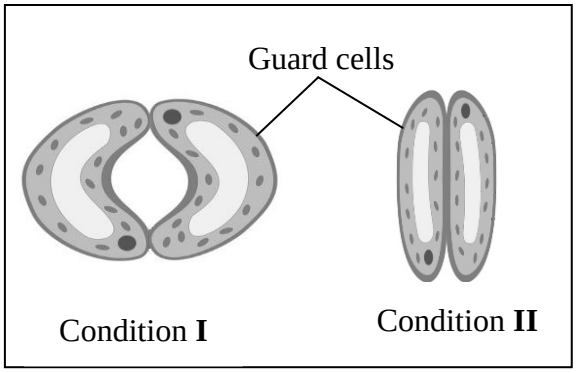


The direction of movement of water molecules across the given cells would be

A. III→II→I
B. III→I→II
C. II→I→III
D. II→III→I

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33. The given diagram represents guard cells of stomata in two different conditions.



Which of the following options is TRUE about conditions **I** and **II** in guard cells?

	Condition I		**Condition II**	
	Concentration of K^+ ions	**Concentration of Water**	**Concentration of K^+ ions**	**Concentration of Water**
A	High	High	Low	Low
B	Low	Low	High	High
C	Low	High	High	Low
D	High	Low	Low	High

34. A characteristic feature of the closed circulatory system is the

A. absence of blood capillaries.
B. regulation of blood pressure.
C. absence of respiratory pigments.
D. flow of haemolymph in cavities.

35. In the human heart, the tricuspid valve closes when

A. aortic pressure exceeds ventricular pressure.
B. left atrial pressure exceeds left ventricular pressure.
C. right ventricular pressure exceeds right atrial pressure.
D. pulmonary trunk pressure exceeds ventricular pressure.

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